CASE STUDY SPECIFICATION & PREDICTIVE FRAMEWORK

Employee Attrition Analysis and Predictive Modeling Using R

A Data-Driven Approach to Improving Employee Retention & Workforce Stability

DOMAIN	PRIMARY TOOL	MODELING ALGORITHM	TARGET OUTCOME
HR Analytics & Intelligence	R / RStudio / Caret	Random Forest Classifier	Proactive Attrition Reduction

CHAPTER 1: INTRODUCTION

1.1 Background

Employee attrition is one of the most significant operational and strategic challenges faced by contemporary organizations. High turnover rates incur substantial direct recruitment and onboarding costs, impair operational productivity, erode institutional knowledge, and diminish overall team morale. Understanding the multi-faceted factors that drive voluntary resignations empowers Human Resource (HR) departments to design targeted, evidence-based retention interventions.

With the advancement of workforce analytics, modern enterprises increasingly leverage employee administrative and engagement data to reveal subtle turnover patterns. Machine learning techniques further enhance HR capabilities, moving beyond historical reporting to actively forecast which employees are at high risk of resigning before notice is submitted. This case study demonstrates how the R programming language was deployed to analyze employee attrition data, identify dominant drivers of turnover, and train a Random Forest classifier capable of pinpointing vulnerable personnel.

1.2 Problem Statement

The organization has experienced a noticeable increase in employee turnover. Management currently lacks structured data insight into:

- Which operational departments suffer the highest attrition rates.
- Which specific job roles are most severely impacted.
- The underlying work environments, compensation patterns, and career dynamics that drive resignations.
- Which individual employees present the highest probability of leaving.
- How predictive analytics can transform reactive HR decision-making into proactive retention strategies.

Without rigorous analytical diagnostic systems, organizational responses remain purely reactive, costly, and ineffective.

1.3 Aim

To analyze employee attrition using statistical and machine learning techniques in R to uncover the key factors influencing turnover and develop an accurate predictive model for proactive employee retention.

1.4 Objectives

- **Data Preparation:** Import, clean, and preprocess organizational employee data into R.
- **Baseline Analytics:** Calculate the baseline organizational attrition rate.
- **Departmental Analysis:** Measure and evaluate attrition variance across organizational departments.
- **Role Profiling:** Evaluate turnover concentrations across distinct job roles.
- **Factor Exploration:** Investigate major drivers, including overtime, career stagnation, salary, and satisfaction.
- **High-Risk Identification:** Define composite risk profiles representing highly vulnerable employees.
- **Predictive Modeling:** Train and evaluate a Random Forest classification model.
- **Strategic Recommendations:** Formulate actionable, data-backed HR retention policies.

1.5 Research Questions

1. What is the baseline employee attrition rate across the entire workforce?

2. Which department experiences the highest employee turnover rate?

3. Which job roles are most vulnerable to employee resignations?

4. How significantly does mandatory or frequent overtime influence turnover?

5. Does measured job satisfaction correlate inversely with employee attrition?

6. How does promotion delay (career stagnation) affect resignation likelihood?

7. Does poor work-life balance serve as a primary catalyst for staff departure?

8. What role does monthly income relative to median compensation play in retention?

9. Can a Random Forest machine learning model accurately predict individual attrition risk?

CHAPTER 2: CASE STUDY DESCRIPTION

Organization Overview

The target organization is a medium-to-large enterprise operating across several core departments, including Human Resources, Sales, and Research & Development (R&D). Despite offering market-competitive baseline salaries, executive leadership observed an alarming upward trend in voluntary employee resignations.

The HR department maintains comprehensive digital employee records capturing demographic details, tenure, compensation, promotion history, work-life balance ratings, overtime logs, environment satisfaction ratings, and job satisfaction evaluations. Management commissioned this data analytics project in R to transform these disparate data assets into strategic workforce intelligence.

Business Problem & Objectives

The primary executive imperative is to transition HR from an administrative reporting unit into an analytical decision center by resolving four vital operational questions:

1. Which employees exhibit a high mathematical probability of resigning in the near term?
2. What specific combination of workplace conditions, compensation tiers, and workload dynamics triggers resignations?
3. How can HR business partners intervene proactively before high-value employees submit their resignations?
4. Which organizational policies require immediate restructuring to foster long-term retention?

CHAPTER 3: METHODOLOGY

Research Method

This project applied a quantitative analytical framework combining descriptive statistical profiling, bivariate relationship testing, feature creation, and supervised machine learning classification.

Dataset Description

The project utilized the HR Employee Attrition Dataset. Key variables analyzed include:

Variable Name	Description & Data Type
`Age`	Continuous numeric: Employee age in years
`Department`	Categorical: HR, Sales, Research & Development

Variable Name	Description & Data Type
`Job Role`	Categorical: Specific job designation
`Monthly Income`	Continuous numeric: Gross monthly earnings
`Job Satisfaction`	Ordinal scale (1–4): Rated job satisfaction level
`Work-Life Balance`	Ordinal scale (1–4): Rated work-life balance quality
`Overtime Status`	Categorical: Regular overtime worked (Yes / No)
`Years Since Last Promotion`	Continuous numeric: Time elapsed since last formal promotion
`Years at Company`	Continuous numeric: Total tenure at current firm
`Environment Satisfaction`	Ordinal scale (1–4): Workplace environment rating
`Number of Companies Worked`	Continuous numeric: Previous employer count
`Attrition Status`	Target Variable: Binary turnover status (Yes / No)

Software and Libraries

The analytical workflow was executed within RStudio using the following key R ecosystem packages:

R Language core | **tidyverse & dplyr** | **ggplot2 visualization** | **randomForest engine** | **caret framework** | **corrplot & RColorBrewer**

Data Preparation

- Dataset imported and verified for structural integrity.
- `Attrition` target encoded into numeric binary indicators (`Yes = 1`, `No = 0`).
- Baseline metrics established: Total Workforce Count, Resigned Employee Count, and Overall Attrition Rate (%).

Data Analysis Dimensions

1. Departmental Attrition Analysis

Grouped employees by department to aggregate total headcounts, total resignations, and departmental attrition percentages.

Purpose: Pinpoint high-turnover operational divisions.

2. Job Role Attrition Analysis

Segmented workforce by specific job titles to calculate localized resignation volumes and rates.

Purpose: Identify occupations vulnerable to market poaching or burnout.

3. Overtime Impact Analysis

Evaluated turnover rates comparing employees performing frequent overtime against non-overtime peers.

Purpose: Determine workload pressure threshold effects on turnover.

4. Job Satisfaction Analysis

Analyzed attrition distribution across ordinal satisfaction levels (1 = Low to 4 = Very High).

Purpose: Measure the inverse correlation between engagement and exit rates.

5. Career Stagnation Analysis

Engineered binary feature `Stagnant` (True if `YearsSinceLastPromotion > 2`).

Purpose: Test if delayed career advancement triggers voluntary departure.

6. Work-Life Balance Analysis

Examined turnover rates categorized across work-life balance scores.

Purpose: Assess whether lifestyle friction increases turnover risk.

7. Income Level Analysis

Binned monthly earnings into `High Income` and `Low Income` relative to median salary.

Purpose: Determine compensation equity impacts on employee retention.

8. High-Risk Profile Composite Identification

Filtered personnel exhibiting compound risk factors: `Overtime == Yes` AND `JobSatisfaction <= 2` AND `YearsSinceLastPromotion > 2`.

Purpose: Identify high-probability resignation candidates for targeted retention.

Predictive Machine Learning Workflow

A supervised **Random Forest Classification Model** was constructed to predict individual attrition risk:

- **Predictor Variables:** Age, Monthly Income, Job Satisfaction, Overtime, Years Since Last Promotion, Work-Life Balance, Environment Satisfaction, Years At Company, Number of Companies Worked.
- **Target Variable:** Binary Attrition (1 / 0).
- **Data Splitting:** Stratified random partitioning into 70% Training set and 30% Testing set.
- **Model Training:** Random Forest model initialized with `ntree = 100` decision trees.
- **Model Evaluation:** Assessed via Classification Accuracy, Confusion Matrix, and Mean Decrease Gini Feature Importance scores.

CHAPTER 4: EXPECTED RESULTS

Based on exploratory data analysis and initial machine learning runs, the project yields the following empirical findings:

- Establishment of the definitive baseline organizational turnover rate.
- Identification of Sales and R&D roles as key attrition hotspots.
- Strong positive correlation between mandatory overtime hours and resignation probability.
- Significant inverse relationship between recorded job satisfaction scores and turnover.
- Confirmation that career stagnation (>2 years without promotion) dramatically increases departure risk.
- Confirmation that low relative monthly income accentuates vulnerability to external recruitment.
- High overall predictive accuracy delivered by the Random Forest classifier in isolating high-risk personnel prior to departure.

CHAPTER 5: BUSINESS VALUE

Implementing this empirical analysis framework delivers substantial tangible and strategic value to executive leadership:

- **Proactive HR Interventions:** Shifts HR strategy from reactive offboarding to early intervention.
- **Recruitment & Onboarding Cost Reduction:** Decreases expensive replacement hiring cycles.
- **Optimized Talent Pipelines:** Enables structured, timely promotion cycles to mitigate stagnation.
- **Productivity Preservation:** Preserves team productivity and protects organizational memory.
- **Data-Driven Workplace Policy:** Guides work-life balance and compensation benchmarking with evidence.

CHAPTER 6: RECOMMENDATIONS

1. Optimize Work-Life Balance & Overtime Policies

Cap mandatory overtime, implement workload rebalancing, and offer flexible remote work schedules to prevent employee burnout.

2. Establish Transparent Career Progression Pathways

Conduct bi-annual promotion reviews to ensure high-performing personnel do not experience career stagnation beyond 24 months.

3. Enhance Employee Engagement & Satisfaction Systems

Deploy pulse engagement surveys to identify emerging workplace concerns and rapidly resolve departmental friction points.

4. Modernize Compensation & Equity Structures

Conduct regular market salary benchmarking to align lower-tier compensation roles with industry standards.

5. Integrate Predictive HR Dashboards

Embed R predictive scripts directly into enterprise HR management systems to flag high-risk employee profiles in real time.

6. Design Targeted Retention Programs for High-Risk Profiles

Establish executive mentorship and stay-interview programs tailored specifically for personnel presenting compound risk indicators.

CHAPTER 7: CONCLUSION

This case study demonstrates the power of applying statistical computing in R and machine learning algorithms to resolve complex Human Resource challenges. By combining descriptive statistical profiling with a Random Forest predictive model, the project successfully isolated key turnover drivers—specifically overtime burden, low job satisfaction, career stagnation, and income level—and established a reliable method for identifying vulnerable employees.

Adopting this data-driven methodology equips HR leadership to make proactive decisions, optimize retention expenditures, foster a healthier work environment, and maintain a resilient workforce.